CUBE-SUM PERMUTATIONS

A006063: final proof map, finite certificates, and mathematical audit

19 October 2026 | Consolidation of pull request #13. This note separates the proved eventual-existence theorem and exact classifications from the still-experimental higher-start involution theorem.

1. Main theorem and dependency map

For every integer n >= 173, there exists a permutation p of {1,...,n} with i+p(i) a perfect cube for every i. Exactly 48 sizes below 173 are admissible; each of the other 124 is excluded by an explicit Hall-deficient set. Therefore A006063(j)=j+124 for all j>=49.

The proof has three logically distinct layers. (I) Establish 82 independent seed sizes using 56 integer-determinant shape instances, a forced-neighbor proof at n=173, and 25 instances of six symbolic block templates. (II) Cubic suffix reflection supplies 26 more smaller sizes and 132 further sizes inside [173,364], covering all 192 target lengths. (III) A completely symbolic interval-overlap induction extends [173,364] to all n>=173.

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56 determinant shapes + 1 forced seed + 25 template seeds = 82 seeds
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60 seeds in [173,364] + 132 reflected lengths = 192 lengths
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2. Universal block construction

Choose positive input block lengths l[1],...,l[t], and reorder the same blocks in the output by a permutation pi. Let A[j] be the total length before input block j, B[j] the total length before output block j, and C[j]=1+A[j]+B[j]+l[j]. The map p(i)=C[j]-i on input block j sends that block bijectively onto its output block. If the C[j] are cubes, the resulting map is a cube-sum permutation.

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M[j,i]=[i<j]+[pos_pi(i)<pos_pi(j)]+[i=j]
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M*l=(k[j]^3-1), l[j]=det(M with column j replaced by d)
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The chord-intersection/Pfaffian argument in the full manuscript shows det(M) lies in {0,1}. Each of the 56 selected root-word/output-order pairs has determinant 1. Exact integer evaluation reconstructs all 784 positive block lengths and verifies the specified total sizes. These selected pairs and positivity conditions remain finite case data: the result is not case-free.

Each certified shape also gives a family under scaling cube roots by h: put v=M^{-1}1 and e=t mod 2. With v[j] in {-1,0,1}, the lengths l[j](h)=l[j]+(h^3-1)(l[j]+v[j]) remain positive. The new total length equals (N+e)h^3-e.

3. Finite base and forced peeling

The 56 determinant pairs contribute 10 seed sizes below 173 and 46 sizes in [173,364]. Six algebraic block-template families contribute 12 and 13 more, respectively. The missing critical seed n=173 has a different, forced-neighbor proof. Totals: 22 lower seeds and 60 target-interval seeds.

In the undirected cube-sum graph, a vertex i having one neighbor j forces p(i)=j and p(j)=i. Removing the pair preserves the count of valid permutations. For n=173 and n=174, repeated forced

removal exhausts the graph; n=175 leaves a 10-vertex loop/edge/odd-cycle core. Hence P(173)=P(174)=1 and P(175)=3.

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P(173)=1, P(174)=1, P(175)=3
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The full 57-shape determinant table and all historical endpoint atlases are redundant alternative certificates. The reduced route needs neither the original 191 augmenting words nor a matching search in its main inference.

4. Reflection closure and interval-overlap avalanche

If G(m) denotes existence at size m and c=m+n+1 is a cube with 0<=m<n, retain the permutation on [1,m] and append p(i)=c-i for i in [m+1,n]. This is a valid permutation at n because the appended suffix maps onto itself. The finite closure yields all 48 admissible sizes below 173 and all integers 173,...,364. In particular sizes 256,...,338 reflect from m=511-n under cube c=512.

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G(m) and c=m+n+1 cube (0<=m<n) => G(n)
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Interval lemma: if every n in [a,b] is good and a cube c satisfies a+b+2<=c<=2*b+2, then every n in [a,c-a-1] is good. Indeed m=c-n-1 lies in [a,b] for every new n. Starting from [173,364], the cube 729 yields [173,555]. Successive cubes (k+1)^3 extend [173,k^3-174] to [173,(k+1)^3-174], k>=9, so the upper endpoints diverge. This infinite tail has no new finite certificates.

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G([a,b]), a+b+2<=cube c<=2*b+2 => G([a,c-a-1])
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Interval coverage is not itself a Hall theorem: at n=65 the sets of possible cube reflections cover all positions but S={3,40,59,64} has N(S)={5,24,61}. A new matching/gluing theorem would be required to remove the finite base.

5. Exact classifications: separate finite certificates

For n=1,...,172, 48 explicit admissible permutations and 124 individual Hall-deficient sets settle membership. These are finite arithmetic certificates; they also establish that 173 is a sharp threshold. The main existence theorem alone does not establish the 124 negative cases.

Let P(n) be the number of cube-sum permutations. Exactly 13 smaller admissible sizes have an alternating-cycle witness; the other 35 positive small sizes have acyclic alternating digraphs and are unique. The 13 multiple-solution sizes are 112,115,116,117,119,124,125,126,127,128,129,130,133.

The second- and third-solution archives provide distinct alternatives for n=175,...,364 and the two special extension lengths 554 and 555. Symbolic reflection then yields P(n)>=3 for n>=175. Thus A096680(j)=j+161 for j>=14. These multiplicity claims require their own finite certificate checks.

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A096680(j)=j+161 for j>=14; P(n)>=3 for all n>=175
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6. Other results and the open four-cube branch

The long research manuscript contains further independent directions: P(236)=512 and no involution at n=236, alternating-cycle gadgets, entropy of reflection trees, taxicab switching, parity-minimal involutions, exponential constructions, and algorithms. They should be read as separate results, not prerequisites of the existence theorem. They have not all received an independent line-by-line audit in this consolidation.

A different four-cube involution program studies 199 cases in [467,665]. It has experimentally positive general-graph matchings; its structural reductions establish three endpoint planar

embeddings, 26 face templates, fixed deletions of 14/10/16 edges to bipartite graphs, and four Phase I correction patterns. A uniform phase-wide Hall proof is NOT currently supplied. This open theorem is unnecessary for the already proved existence of arbitrary cube-sum permutations for n>=173.

7. Mathematical correction and independent audit

Correction in main.tex: the minimal number of additional seeds under the strict reflection-only rule, assuming the 48 positive small sizes and empty seed already known, is 60, not 57. Its proof explicitly counts 44 seeds in [173,255] and 16 in [339,364], none in [256,338]. The number 57 described an older, different determinant-shape certificate, not this restricted minimum.

Independent exact checks found zero failures for: all 56 determinants and 784 positive generated lengths, the forced peeling cores at 173/174/175, all 48 small positive permutations, all 124 Hall-deficient sets (strict deficits 1 through 6), all 13 smaller alternating-cycle witnesses, and acyclicity of the remaining 35 smaller positive cases. The audit does not independently certify every other ancillary theorem or all finite large-n multiplicity rows.

The two previous GitHub Actions runs on head 47d3f5a returned action_required, not confirmed success. Independent peer review and literature-priority checking remain open tasks; absence of a completed CI run is not evidence for or against the mathematics.

8. Where to read the full proof and reproduce it

Main proof order: COMPRESSED_FINITE_PROOF.md; UNIMODULAR_BLOCK_FACTORY.md; FORCED_REFLECTION_PEELING.md; the six block-template notes; INTERVAL_OVERLAP_AVALANCHE.md. Exact finite audit: verify_56_plus_forcing.py. Full membership and multiplicity audit: verify_extended.py. High-start branch: FOUR_CUBE_PHASES.md, SMALL_BIPARTITE_CORE.md, PHASE_I_FOUR_HALL_PATTERNS.md.

Repository: github.com/dongxuelian2/oeis-papers, pull request #13, papers/A006063/. The principal finite root-order parameter table is certificates/finite_56_cubic_order_shapes.txt. For the extended result inventory and evidence boundaries, see FINAL_READER_GUIDE.md.

Conclusion: the n>=173 existence claim is a complete explicit finite-certificate proof followed by a symbolic infinite theorem, but it is not a case-free proof. The higher-start uniform four-cube involution theorem remains open.